

20. Find the curvature at $t = \pi$ of the cycloid that is described by

$$\mathbf{r}(t) = a(t - \sin t)\mathbf{i} + a(1 - \cos t)\mathbf{j}, \quad a > 0.$$

21. Let C be a plane curve traced by $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$, where f and g have second derivatives. Show that the curvature at a point is given by

$$\kappa = \frac{|f'(t)g''(t) - g'(t)f''(t)|}{([f'(t)]^2 + [g'(t)]^2)^{3/2}}.$$

22. Show that if $y = F(x)$, the formula for κ in Problem 21 reduces to

$$\kappa = \frac{|F''(x)|}{[1 + (F'(x))^2]^{3/2}}.$$

In Problems 23 and 24, use the result of Problem 22 to find the curvature and radius of curvature of the curve at the indicated points. Decide at which point the curve is “sharper.”

23. $y = x^2$; $(0, 0)$, $(1, 1)$
 24. $y = x^3$; $(-1, -1)$, $(\frac{1}{2}, \frac{1}{8})$

Discussion Problems

25. Discuss the curvature near a point of inflection of $y = F(x)$.
 26. Show that $\|\mathbf{a}(t)\|^2 = a_N^2 + a_T^2$.

9.4 Partial Derivatives

Introduction In this section we consider functions of two or more variables and how to find the instantaneous rate of change—that is, the derivative—of such functions with respect to each variable.

Functions of Two Variables Recall from calculus that a **function of two variables** is a rule of correspondence that assigns to each ordered pair of real numbers (x, y) of a subset of the xy -plane one and only one number z in the set R of real numbers. The set of ordered pairs (x, y) is called the **domain** of the function and the set of corresponding values of z is called the **range**. A function of two variables is usually written $z = f(x, y)$. The variables x and y are called the **independent variables** of the function, and z is called the **dependent variable**. The graph of a function $z = f(x, y)$ is a surface in 3-space. See **FIGURE 9.4.1**.

Level Curves For a function $z = f(x, y)$, the curves defined by $f(x, y) = c$, for suitable c , are called the **level curves** of f . The word *level* arises from the fact that we can interpret the equation $f(x, y) = c$ as the projection onto the xy -plane of the curve of intersection, or trace, of $z = f(x, y)$ and the (horizontal or level) plane $z = c$. See **FIGURE 9.4.2**.

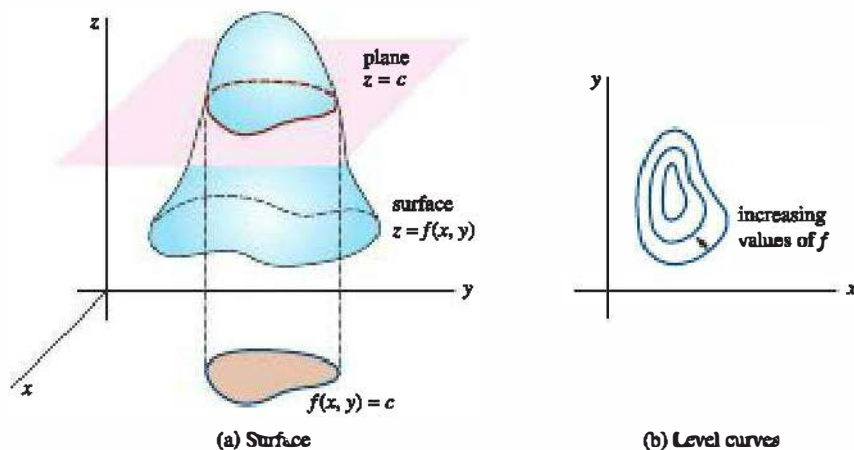


FIGURE 9.4.2 Surface and level curves

EXAMPLE 1 Level Curves

The level curves of the function $f(x, y) = y^2 - x^2$ are defined by $y^2 - x^2 = c$. As shown in **FIGURE 9.4.3**, when $c > 0$ or $c < 0$, a member of this family of curves is a hyperbola. For $c = 0$, we obtain the lines $y = x$ and $y = -x$.

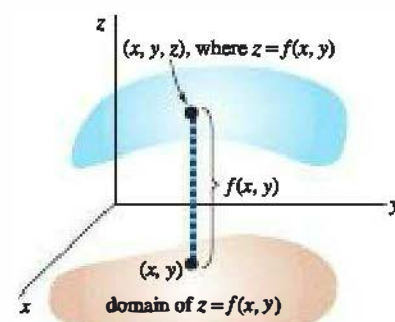


FIGURE 9.4.1 Function of two variables

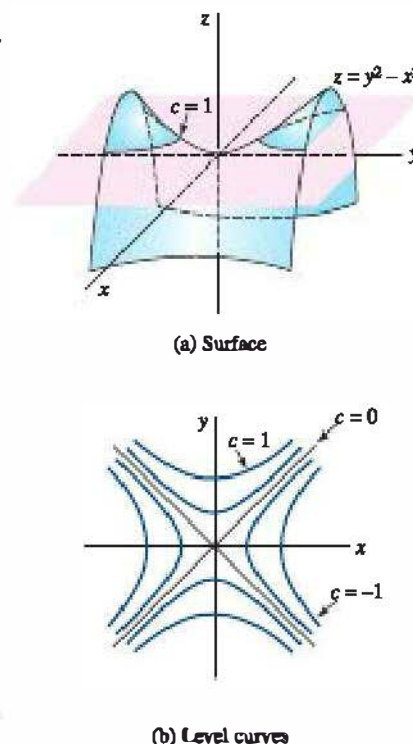


FIGURE 9.4.3 Surface and level curves in Example 1

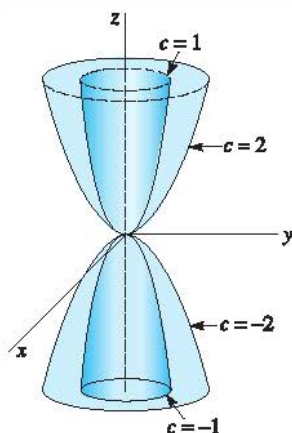


FIGURE 9.4.4 Level surfaces in Example 2

Functions of Three or More Variables Functions of three or more variables are defined analogously to functions of two variables. For example, a **function of three variables** is a rule of correspondence that assigns to each ordered triple of real numbers (x, y, z) of a subset of 3-space one and only one number w in the set R of real numbers. We write $w = F(x, y, z)$.

Level Surfaces Although we cannot draw a graph of a function of three variables $w = F(x, y, z)$, we can draw the surfaces defined by $F(x, y, z) = c$ for suitable values of c . These surfaces are called **level surfaces**. This is an unfortunate, though standard, choice of words, since level surfaces are usually not level.

EXAMPLE 2 Level Surfaces

Describe the level surfaces of the function $F(x, y, z) = (x^2 + y^2)/z$.

SOLUTION For $c \neq 0$ the level surfaces are given by

$$\frac{x^2 + y^2}{z} = c \quad \text{or} \quad x^2 + y^2 = cz.$$

A few members of this family of paraboloids are shown in FIGURE 9.4.4. ≡

Partial Derivatives The derivative of a function of one variable $y = f(x)$ is given by

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

In exactly the same manner, we can define a derivative of a function of two variables with respect to each variable. If $z = f(x, y)$, then the **partial derivative with respect to x** is

$$\frac{\partial z}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x} \quad (1)$$

and the **partial derivative with respect to y** is

$$\frac{\partial z}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}, \quad (2)$$

provided each limit exists.

In (1) the variable y does not change in the limiting process; that is, y is held fixed. Similarly, in (2) the variable x is held fixed. The two partial derivatives (1) and (2) then represent the *rates of change* of f with respect to x and y , respectively. On a practical level:

To compute $\partial z/\partial x$, use the laws of ordinary differentiation while treating y as a constant.

To compute $\partial z/\partial y$, use the laws of ordinary differentiation while treating x as a constant.

EXAMPLE 3 Partial Derivatives

If $z = 4x^3y^2 - 4x^2 + y^6 + 1$, find $\partial z/\partial x$ and $\partial z/\partial y$.

SOLUTION We hold y fixed and treat constants in the usual manner. Thus,

$$\frac{\partial z}{\partial x} = 12x^2y^2 - 8x.$$

By treating x as a constant, we obtain

$$\frac{\partial z}{\partial y} = 8x^3y + 6y^5. \quad \equiv$$

Alternative Symbols The partial derivatives $\partial z/\partial x$ and $\partial z/\partial y$ are often represented by alternative symbols. If $z = f(x, y)$, then

$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial x} = z_x = f_x \quad \text{and} \quad \frac{\partial z}{\partial y} = \frac{\partial f}{\partial y} = z_y = f_y.$$

□ **Higher-Order and Mixed Derivatives** For a function of two variables $z = f(x, y)$, the partial derivatives $\partial z/\partial x$ and $\partial z/\partial y$ are themselves functions of x and y . Consequently, we can compute **second** and higher **partial derivatives**. Indeed, we can find the partial derivative of $\partial z/\partial x$ with respect to y , and the partial derivative of $\partial z/\partial y$ with respect to x . The latter types of partial derivatives are called **mixed partial derivatives**. In summary, for $z = f(x, y)$:

Second-order partial derivatives:

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) \quad \text{and} \quad \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right).$$

Third-order partial derivatives:

$$\frac{\partial^3 z}{\partial x^3} = \frac{\partial}{\partial x} \left(\frac{\partial^2 z}{\partial x^2} \right) \quad \text{and} \quad \frac{\partial^3 z}{\partial y^3} = \frac{\partial}{\partial y} \left(\frac{\partial^2 z}{\partial y^2} \right).$$

Mixed second-order partial derivatives:

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) \quad \text{and} \quad \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right).$$

□ **Alternative Symbols** The second- and third-order partial derivatives are denoted by f_{xx} , f_{yy} , f_{xxx} , and so on. The subscript notation for mixed second partial derivatives is f_{xy} or f_{yx} . Note that

$$f_{xy} = (f_x)_y = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial^2 z}{\partial y \partial x} \quad \text{and} \quad f_{yx} = \frac{\partial^2 z}{\partial x \partial y}.$$

Although we shall not prove it, if a function f has continuous second partial derivatives, then the order in which a mixed second partial derivative is done is irrelevant; that is,

$$f_{xy} = f_{yx}. \quad (3)$$

□ **Functions of Three or More Variables** The rates of change of a function of three variables $w = F(x, y, z)$ in the x , y , and z directions are $\partial w/\partial x$, $\partial w/\partial y$, and $\partial w/\partial z$, respectively. To compute, say, $\partial w/\partial x$, we differentiate with respect to x in the usual manner while holding *both* y and z constant. In this manner we extend the process of partial differentiation to functions of any number of variables.

EXAMPLE 4 Partial Derivatives

If $F(x, y, t) = e^{-3\pi t} \cos 4x \sin 6y$, then the partial derivatives with respect to x , y , and t are, in turn,

$$F_x(x, y, t) = -4e^{-3\pi t} \sin 4x \sin 6y,$$

$$F_y(x, y, t) = 6e^{-3\pi t} \cos 4x \cos 6y,$$

$$F_t(x, y, t) = -3\pi e^{-3\pi t} \cos 4x \sin 6y. \quad \equiv$$

□ **Chain Rule** The Chain Rule for functions of one variable states that if $y = f(u)$ is a differentiable function of u , and $u = g(x)$ is a differentiable function of x , then the derivative of the composite function is

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}. \quad (4)$$

For a composite function of two variables $z = f(u, v)$, where $u = g(x, y)$ and $v = h(x, y)$, we would naturally expect *two* formulas analogous to (4), since we can compute both $\partial z/\partial x$ and $\partial z/\partial y$. The **Chain Rule** for functions of two variables is summarized as follows:

Theorem 9.4.1 Chain Rule

If $z = f(u, v)$ is differentiable and $u = g(x, y)$ and $v = h(x, y)$ have continuous first partial derivatives, then

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x}, \quad \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y}. \quad (5)$$

EXAMPLE 5 Chain Rule

If $z = u^2 - v^3$ and $u = e^{2x-3y}$, $v = \sin(x^2 - y^2)$, find $\partial z/\partial x$ and $\partial z/\partial y$.

SOLUTION Since $\partial z/\partial u = 2u$ and $\partial z/\partial v = -3v^2$, it follows from (5) that

$$\frac{\partial z}{\partial x} = 2u(2e^{2x-3y}) - 3v^2[2x \cos(x^2 - y^2)] = 4ue^{2x-3y} - 6xv^2 \cos(x^2 - y^2) \quad (6)$$

$$\frac{\partial z}{\partial y} = 2u(-3e^{2x-3y}) - 3v^2[(-2y) \cos(x^2 - y^2)] = -6ue^{2x-3y} + 6yv^2 \cos(x^2 - y^2). \quad (7) \equiv$$

Of course, in Example 5 we could substitute the expressions for u and v in the original function and then find the partial derivatives directly. In the same manner, the answers (6) and (7) can be expressed in terms of x and y .

Special Case If $z = f(u, v)$ is differentiable and $u = g(t)$ and $v = h(t)$ are differentiable functions of a single variable t , then Theorem 9.4.1 implies that the ordinary derivative dz/dt is

$$\frac{dz}{dt} = \frac{\partial z}{\partial u} \frac{du}{dt} + \frac{\partial z}{\partial v} \frac{dv}{dt}. \quad (8)$$

Generalizations The results given in (5) and (8) immediately generalize to any number of variables. If $z = f(u_1, u_2, \dots, u_n)$ and each of the variables $u_1, u_2, u_3, \dots, u_n$ are functions of $x_1, x_2, \dots, x_k$, then under the same assumptions as in Theorem 9.4.1 we have

$$\frac{\partial z}{\partial x_i} = \frac{\partial z}{\partial u_1} \frac{\partial u_1}{\partial x_i} + \frac{\partial z}{\partial u_2} \frac{\partial u_2}{\partial x_i} + \dots + \frac{\partial z}{\partial u_n} \frac{\partial u_n}{\partial x_i}, \quad (9)$$

where $i = 1, 2, \dots, k$. Similarly, if the u_i , $i = 1, \dots, n$, are differentiable functions of a single variable t , then

$$\frac{dz}{dt} = \frac{\partial z}{\partial u_1} \frac{du_1}{dt} + \frac{\partial z}{\partial u_2} \frac{du_2}{dt} + \dots + \frac{\partial z}{\partial u_n} \frac{du_n}{dt}. \quad (10)$$

Tree Diagrams The results in (5) can be memorized in terms of a **tree diagram**. The dots in the first diagram in **FIGURE 9.4.5(a)** indicate the fact that z depends on u and v ; u and v depend, in turn, on x and y . To compute $\partial z/\partial x$ for example, we read the diagram vertically downward starting from z and following the two blue polygonal paths leading to x , multiply the partial derivatives on each path, and then add the products. The result given in (8) is represented by the second tree diagram in Figure 9.4.5(b).

We shall use tree diagrams in the next two examples to illustrate special cases of (9) and (10).

EXAMPLE 6 Using Tree Diagrams

If $r = x^2 + y^5 z^3$ and $x = uve^{2s}$, $y = u^2 - v^2 s$, $z = \sin(uvs^2)$, find $\partial r/\partial s$.

SOLUTION From the blue and green paths in the tree diagram in Figure 9.4.5(c) we obtain

$$\begin{aligned} \frac{\partial r}{\partial s} &= \frac{\partial r}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial r}{\partial y} \frac{\partial y}{\partial s} + \frac{\partial r}{\partial z} \frac{\partial z}{\partial s} \\ &= 2x(2uve^{2s}) + 5y^4 z^3(-v^2) + 3y^5 z^2(2uvs \cos(uvs^2)). \quad \equiv \end{aligned}$$

EXAMPLE 7 Using Tree Diagrams

If $z = u^2 v^3 w^4$ and $u = t^2$, $v = 5t - 8$, $w = t^3 + t$, find dz/dt .

SOLUTION In this case the tree diagram in Figure 9.4.5(d) indicates that

$$\begin{aligned} \frac{dz}{dt} &= \frac{\partial z}{\partial u} \frac{du}{dt} + \frac{\partial z}{\partial v} \frac{dv}{dt} + \frac{\partial z}{\partial w} \frac{dw}{dt} \\ &= 2uv^3w^4(2t) + 3u^2v^2w^4(5) + 4u^2v^3w^3(3t^2 + 1). \end{aligned}$$

ALTERNATIVE SOLUTION Differentiate $z = t^4(5t - 8)^3(t^3 + t)^4$ by the Product Rule. $\equiv$

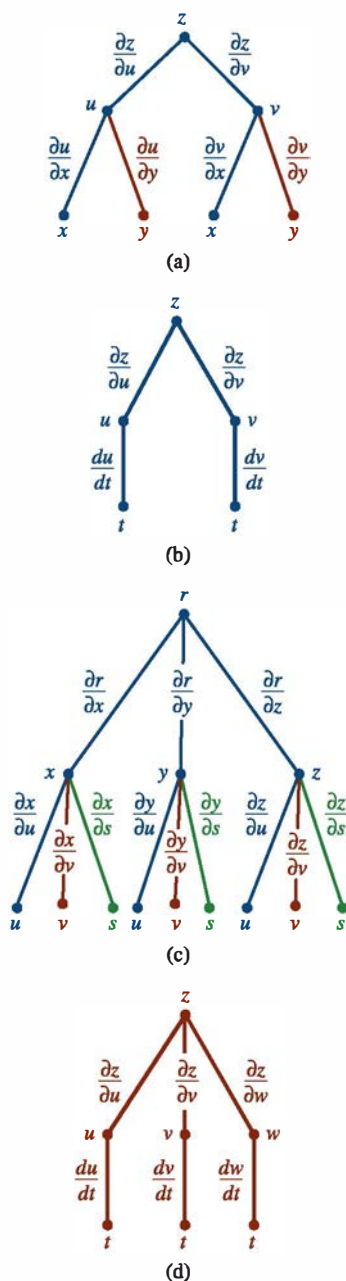


FIGURE 9.4.5 Tree diagrams

Remarks

If $w = F(x, y, z)$ has continuous partial derivatives of any order, then analogous to (3), the mixed partial derivatives are equal:

$$F_{xyz} = F_{yzx} = F_{zyx}, \quad F_{xxy} = F_{yxx} = F_{xyx},$$

and so on.

9.4 Exercises

Answers to selected odd-numbered problems begin on page ANS-21.

In Problems 1–6, sketch some of the level curves associated with the given function.

1. $f(x, y) = x + 2y$
2. $f(x, y) = y^2 - x$
3. $f(x, y) = \sqrt{x^2 - y^2 - 1}$
4. $f(x, y) = \sqrt{36 - 4x^2 - 9y^2}$
5. $f(x, y) = e^{y-x^2}$
6. $f(x, y) = \tan^{-1}(y - x)$

In Problems 7–10, describe the level surfaces but do not graph.

7. $F(x, y, z) = \frac{x^2}{9} + \frac{z^2}{4}$
8. $F(x, y, z) = x^2 + y^2 + z^2$
9. $F(x, y, z) = x^2 + 3y^2 + 6z^2$
10. $F(x, y, z) = 4y - 2z + 1$
11. Graph some of the level surfaces associated with $F(x, y, z) = x^2 + y^2 - z^2$ for $c = 0$, $c > 0$, and $c < 0$.
12. Given that

$$F(x, y, z) = \frac{x^2}{16} + \frac{y^2}{4} + \frac{z^2}{9},$$

find the x -, y -, and z -intercepts of the level surface that passes through $(-4, 2, -3)$.

In Problems 13–32, find the first partial derivatives of the given function.

13. $z = x^2 - xy^2 + 4y^5$
14. $z = -x^3 + 6x^2y^3 + 5y^2$
15. $z = 5x^4y^3 - x^2y^6 + 6x^5 - 4y$
16. $z = \tan(x^3y^2)$
17. $z = \frac{4\sqrt{x}}{3y^2 + 1}$
18. $z = 4x^3 - 5x^2 + 8x$
19. $z = (x^3 - y^2)^{-1}$
20. $z = (-x^4 + 7y^2 + 3y)^6$
21. $z = \cos^2 5x + \sin^2 5y$
22. $z = e^{x^2 \tan^{-1} y^2}$
23. $f(x, y) = xe^{xy}$
24. $f(\theta, \phi) = \phi^2 \sin \frac{\theta}{\phi}$
25. $f(x, y) = \frac{3x - y}{x + 2y}$
26. $f(x, y) = \frac{xy}{(x^2 - y^2)^2}$
27. $g(u, v) = \ln(4u^2 + 5v^3)$
28. $h(r, s) = \frac{\sqrt{r}}{s} - \frac{\sqrt{s}}{r}$
29. $w = 2\sqrt{xy} - ye^{y/z}$
30. $w = xy \ln(xz)$
31. $F(u, v, x, t) = u^2w^2 - uv^3 + vw \cos(ut^2) + (2x^2t)^4$
32. $G(p, q, r, s) = (p^2q^3)^{r^4s^5}$

In Problems 33 and 34, verify that the given function satisfies Laplace's equation:

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0.$$

33. $z = \ln(x^2 + y^2)$
34. $z = e^{x^2 - y^2} \cos 2xy$

In Problems 35 and 36 verify that the given function satisfies the wave equation:

$$a^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}.$$

35. $u = \cos at \sin x$
36. $u = \cos(x + at) + \sin(x - at)$
37. The molecular concentration $C(x, t)$ of a liquid is given by $C(x, t) = t^{-1/2} e^{-x^2/4t}$. Verify that this function satisfies the diffusion equation:

$$\frac{k}{4} \frac{\partial^2 C}{\partial x^2} = \frac{\partial C}{\partial t}.$$

38. The pressure P exerted by an enclosed ideal gas is given by $P = k(T/V)$, where k is a constant, T is temperature, and V is volume. Find:
 - (a) the rate of change of P with respect to V ,
 - (b) the rate of change of V with respect to T , and
 - (c) the rate of change of T with respect to P .

In Problems 39–48, use the Chain Rule to find the indicated partial derivatives.

39. $z = e^{uv^2}$; $u = x^3$, $v = x - y^2$; $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$
40. $z = u^2 \cos 4v$; $u = x^2y^3$, $v = x^3 + y^3$; $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$
41. $z = 4x - 5y^2$; $x = u^4 - 8v^3$, $y = (2u - v)^2$; $\frac{\partial z}{\partial u}$, $\frac{\partial z}{\partial v}$
42. $z = \frac{x - y}{x + y}$; $x = \frac{u}{v}$, $y = \frac{v^2}{u}$; $\frac{\partial z}{\partial u}$, $\frac{\partial z}{\partial v}$
43. $w = (u^2 + v^2)^{3/2}$; $u = e^{-t} \sin \theta$, $v = e^{-t} \cos \theta$; $\frac{\partial w}{\partial t}$, $\frac{\partial w}{\partial \theta}$
44. $w = \tan^{-1} \sqrt{uv}$; $u = r^2 - s^2$, $v = r^2 s^2$; $\frac{\partial w}{\partial r}$, $\frac{\partial w}{\partial s}$
45. $R = rs^2t^4$; $r = ue^{v^2}$, $s = ve^{-u^2}$, $t = e^{u^2v^2}$; $\frac{\partial R}{\partial u}$, $\frac{\partial R}{\partial v}$
46. $Q = \ln(pqr)$; $p = t^2 \sin^{-1} x$, $q = \frac{x}{t^2}$, $r = \tan^{-1} \frac{x}{t}$; $\frac{\partial Q}{\partial x}$, $\frac{\partial Q}{\partial t}$